

THE CONVOLUTION OF THE NORMAL AND LOGNORMAL DISTRIBUTIONS

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Key words: Calibration; conditional distribution; estimation.

Summary: Let Z be a three-parameter lognormal variate, U a normal with zero mean and define $X = Z + U$. The marginal distribution of X is then the convolution of the lognormal with the normal - a distribution we will abbreviate to LNN. Expressions for the density and distribution function of the LNN are given, and its properties sketched. Maximum likelihood and method of moments estimators of the parameters of the LNN are given. Calibration requires the conditional distribution of X given Z . This is derived, along with the calibration curve $E(X|Z)$ and its standard error function. An application to X-ray fluorescence counting of in situ gold grades is discussed.

1. INTRODUCTION

The lognormal distribution has found application in many problem areas (Aitchison and Brown 1957) in a variety of fields ranging from biology to economics. It is particularly useful in mining, as the De Wijsian model of ore deposits (Matheron 1960) suggests that it may apply to any mineral present in low concentrations, and this has been verified experimentally for oil and many minerals (Krige 1978, Harbaugh and Ducastaing 1981). Its use for gold deposits was pioneered by Sichel (1947), who in extensive subsequent research and writings has developed much of the corpus of knowledge on inference on the parameters of the lognormal and derived quantities such as the arithmetic mean. See for example Sichel (1987) for some recent developments in lognormal theory.

The three-parameter lognormal was introduced to geology by Krige (1960), for modelling gold and uranium grades and is now widely regarded as the 'natural' parametric model for low-concentration deposits.

It has been the standard practice when studying lognormal data to ignore any measurement error there might be in observing the lognormal variable. For many applications this assumption is justified by the high precision of measurement relative to the true variability. Recently, however, there has been a growing interest in the mining community in measuring lognormally distributed quantities with instruments which give quick and inexpensive readings having substantial random measurement errors.

The project which originally gave rise to this research is a measuring instrument of this type (Rolle 1979) in which an exposed mine face is irradiated with high energy electromagnetic sources causing each element in the ore (including those of economic interest) to fluoresce at specific wavelengths. The counts C say in the valuable element's 'channel' provide an estimate of the concentration (or 'grade') of the element, but this has first to be corrected for background radiation, which is done by subtracting a suitable multiple of the counts B say obtained in a 'background' channel. Thus the final reading is given by $C - aB$. While this reading is unbiased for the true concentration of the valuable element, it has a random error with a large standard deviation – often for example exceeding the mean of the underlying quantity. This led to the perception of a need for the distributions associated with a three-parameter lognormal quantity with substantial measurement error. We chose to model this measurement error with a homoscedastic normal distribution. While on the face of it the corrected reading in any channel should be modelled by a difference of scaled Poisson rather than normal distributions, for low-concentration analytes the 'background correction' typically involves many thousands of counts and leads to effectively normal homoscedastic measurement errors as has been verified both theoretically and empirically (see for example Magri 1983).

The same statistical framework of an underlying lognormal quantity measured with approximately normal homoscedastic error is plausible in

some other contexts as well. Also in the context of gold mining, suggestions have been made of using the empirically observed co-occurrence of gold and uranium as the basis for estimating gold content with scintillation counters measuring uranium daughter products. In the environmental arena, the lognormal is a natural model for low concentration pollutants. While so-called Type I assays, which measure such pollutants directly, are amenable to the traditional approach of ignoring measurement error, there is growing interest in the use of Type II and Type III indirect assays. These latter types of assay are intended to reduce the enormous cost of assays for a wide array of (particularly organic) pollutants. A Type II assay measures the total concentration of a group of related analytes, while a Type III assay substitutes a measure of a correlated chemical. Both Type II and III assays then require prediction of the individual pollutants of interest, and involve substantial random variability. It is possible (though not automatic) for the same measurement framework of an underlying lognormal quantity with substantial homoscedastic normal measurement error to arise in these contexts too.

Several statistical problems arise within this framework. One is the estimation of the parameters of the underlying lognormal distribution using the LNN readings. Another is the calibration problem – conditional on the LNN reading, determining the distribution of the true grade; and derived from this the calibration curve and its standard error. These problems will be addressed below.

2. DERIVATION OF THE LNN DISTRIBUTION

Let us suppose that the underlying quantity of interest Z follows a three-parameter lognormal distribution (3PLND):

$$\ln(Z+\alpha) \sim N(\xi, \sigma^2).$$

We assume that X is a measurement of Z with a normally distributed random measurement error U –

$$X = Z + U, \text{ where } U \sim N(0, \tau^2).$$

so that conditionally on Z ,

$$X \sim N(Z, \tau^2),$$

and the measurement X of Z is conditionally unbiased with constant variance τ^2 . Then X follows the lognormal-normal convolution (LNN) distribution.

In the development following we will assume τ known. This is the case in the problem motivating the study (where τ can be computed from the theory of Poisson counting statistics), and pertains in many potential applications of the LNN. Where τ is not known, it would be best measured (if possible) by repeat measures X of the same Z , with its estimation directly from the X readings a poor third choice. For this eventuality, however, we do provide the necessary derivatives involving τ^2 as well as those involving the other generally unknown parameters.

Define the two-parameter lognormal (2PLND) quantity $Y = Z + \alpha$. Then the joint distribution of X and Y is easily seen to be

$$\begin{aligned} f(x, y) &= f(y)f(x|y) \\ &= \frac{1}{2\pi\sigma\tau y} \exp -\frac{1}{2} \left[\frac{(\ln y - \xi)^2}{\sigma^2} + \frac{(x - y + \alpha)^2}{\tau^2} \right]. \end{aligned}$$

The marginal density and the cumulative distribution function of X are then

$$(1) \quad g(x|\alpha, \tau, \xi, \sigma) = \int_{y=0}^{\infty} f(x, y) dy$$

$$= \int_{y=0}^{\infty} \frac{1}{2\pi\sigma\tau y} \exp -\frac{1}{2} \left[\frac{(\ln y - \xi)^2}{\sigma^2} + \frac{(x - y + \alpha)^2}{\tau^2} \right] dy$$

$$\begin{aligned}
 G(x|\alpha, \tau, \xi, \sigma) &= \int_{y=0}^{\infty} f(y) \Pr[X \leq x | Y = y] dy \\
 &= \int_{y=0}^{\infty} \Phi\left[\frac{x - y + \alpha}{\tau}\right] \frac{1}{\sqrt{(2\pi)\sigma y}} \exp\left[-\frac{1}{2}\left[\frac{(\ln y - \xi)^2}{\sigma^2}\right]\right] dy
 \end{aligned}$$

where $\Phi(\cdot)$ denotes the standard normal distribution function.

Alternative and computationally more convenient expressions may be obtained by making the transformation $t = \ln y$. This gives:

$$(2) \quad g(x|\alpha, \tau, \xi, \sigma) = \int_{t=-\infty}^{\infty} \frac{1}{2\pi\sigma\tau} \exp\left[-\frac{1}{2}\left[\frac{(t - \xi)^2}{\sigma^2} + \frac{(x - e^t + \alpha)^2}{\tau^2}\right]\right] dt$$

$$G(x|\alpha, \tau, \xi, \sigma) = \int_{t=-\infty}^{\infty} \Phi\left[\frac{x - e^t + \alpha}{\tau}\right] \frac{1}{\sqrt{(2\pi)\sigma}} \exp\left[-\frac{1}{2}\left[\frac{(t - \xi)^2}{\sigma^2}\right]\right] dt$$

Neither representation gives a standard well-known integral, and so both must be evaluated numerically. The details of this are discussed in section 6.

3. PROPERTIES OF THE LNN DISTRIBUTION

The canonical form

The parameters α and τ are location and scale parameters for the LNN. Specifically, writing $X' = aX + b$, it is easily seen that X' has the density $g(x'|a\alpha - b, a\tau, \xi + \ln a, \sigma)$. In particular, $X' = (X + \alpha)/\tau$ has the 'canonical' density of the LNN $g(x'|0, 1, \xi - \ln \tau, \sigma)$. From this, we obtain

$$g(x|\alpha, \tau, \xi, \sigma) = g(\{x + \alpha\}/\tau | 0, 1, \xi - \ln \tau, \sigma)/\tau$$

$$G(x|\alpha, \tau, \xi, \sigma) = G(\{x + \alpha\}/\tau | 0, 1, \xi - \ln \tau, \sigma)$$

so that only the standard canonical form $g(x|0, 1, \xi, \sigma)$ need be studied. The density and cumulative distribution function of this canonical form are

$$(3) \quad g(x|0, 1, \xi, \sigma) = \int_{t=-\infty}^{\infty} \frac{1}{2\pi\sigma} \exp - \frac{1}{2} \left[\frac{(t - \xi)^2}{\sigma^2} + (x - e^t)^2 \right] dt$$

$$G(x|0, 1, \xi, \sigma) = \int_{t=-\infty}^{\infty} \Phi(x - e^t) \frac{1}{\sqrt{(2\pi)\sigma}} \exp - \frac{1}{2} \left[\frac{(t - \xi)^2}{\sigma^2} \right] dt$$

Moments

The moments of the marginal distribution of X may be inferred directly from the definition. Since $X = Z + U$, where $U \sim N(0, \tau^2)$ and is independent of Z ,

$$(4) \quad \begin{aligned} E(X^k) &= E(Z + U)^k \\ &= \sum_{j=0}^k \binom{k}{j} E(Z^j) E(U^{k-j}) \end{aligned}$$

This expression is easily evaluated using the known moments of lognormal and of normal variates. In particular, if we transform to the canonical variate $X' = (X + \alpha)/\tau$ and write $\xi' = \xi - \ln \tau$, the first four moments of X' are given by

$$(5) \quad \begin{aligned} E(X') &= \exp(\xi' + \frac{1}{2}\sigma^2) \\ E(X'^2) &= 1 + \exp(2\xi' + 2\sigma^2) \\ E(X'^3) &= \exp(\xi' + \frac{1}{2}\sigma^2) \{3 + \exp(2\xi' + 4\sigma^2)\} \\ E(X'^4) &= 3 + 6 \exp(2\xi' + 2\sigma^2) + \exp(4\xi' + 8\sigma^2) \end{aligned}$$

On the original scale, the mean and the variance of X are given by:

$$E(X) = \alpha + \exp(\xi + \frac{1}{2}\sigma^2), \quad \text{var}(X) = \tau^2 + \exp(2\xi + \sigma^2)\{(\exp(\sigma^2) - 1)\}.$$

Temporarily defining

$\eta = \exp(\xi' + \frac{1}{2}\sigma^2)$, $\gamma = \exp(\sigma^2)$, we may also write the normalized third moment of X as

$$\frac{E\{X - E(X)\}^3}{[E\{X - E(X)\}^2]^{3/2}} = \frac{\eta^3(\gamma^3 - 3\gamma + 2)}{[1 + \eta^2(\gamma - 1)]^{3/2}}.$$

Note that the sequence of moments of X increases more rapidly than that of the underlying lognormally distributed Z , and that the distribution of X , like the 3PLND, is too heavy-tailed to be determined by its moments.

Asymptotics

It is evident from the definition of X that its extreme right tail is that of the underlying 3PLND variable Z as $x \rightarrow \infty$,

$$G(x | \alpha, \tau, \xi, \sigma) \rightarrow \Phi\{(\ln\{x + \alpha\} - \xi)/\sigma\}.$$

In particular, on lognormal probability paper, the LNN data will give a straight line out to the far right. The key determinant of the length of this asymptotically straight segment is the relative sizes of the variances of Z and U . If the variance of Z is large relative to τ^2 , then the straight segment will tend to be long. Conversely if τ^2 is a substantial portion of the total variance, then the probability plot may be curved over the whole range plotted. Provided the linear segment is visually clear and long enough, its intercept and slope provide quick graphic estimates of the underlying lognormal parameters ξ and σ .

The left tail behaviour is less simple. Expression 3 shows that

$$G(x|0,1, \xi, \sigma) < \Phi(x)$$

with asymptotic equality if ξ is large negative (under which circumstances the lognormal part of X is negligible and X consists essentially of just the measurement noise). If σ is near zero, then G approximates $\Phi[x - \exp(\xi)]$. In the more useful case that ξ is positive and σ appreciable, then the left tail behaviour is not particularly straightforward.

Also interesting is the behaviour for extreme values of the parameters. It is well known that as $\sigma \rightarrow 0$ the lognormal approaches the normal (see for example Klimko et al. 1975 or Kötzt, 1973). Any normal distribution, $N(\mu, \gamma^2)$ say, can be obtained as the degenerate limiting case of the 3PLND letting $\alpha \rightarrow \infty$ and $\sigma \rightarrow 0$ while maintaining the moment equations

$$(6) \quad \alpha + \exp(\xi + \frac{1}{2}\sigma^2) = \mu, \quad \exp(2\xi + \sigma^2)[\exp(\sigma^2) - 1] = \gamma^2.$$

Since the distribution of Z approaches the normal under certain circumstances, then so also must that of the LNN.

The fact that the normal distribution is a special but degenerate limiting case of the LNN has important implications for parameter estimation, as we shall see below.

4. ESTIMATION OF THE PARAMETERS

Apart from the parameter τ , which is assumed known, the distribution g has the three parameters α , ξ and σ whose values must be established before any computations can be made. Let us suppose therefore that we have a sample $X_1, \dots, X_n$ available for this estimation.

Method of maximum likelihood

In the defining equation of $g(x|\alpha, \tau, \xi, \sigma)$, let

$$A_i(t) = \frac{1}{2\pi\sigma\tau} \exp - \frac{1}{2} \left[\frac{(t - \xi)^2}{\sigma^2} + \frac{(X_i - e^t + \alpha)^2}{\tau^2} \right]$$

Using the Macsyma symbolic computation system (Mathlab Group 1983) the first and second derivatives of $g(X_i|\alpha, \tau, \xi, \sigma)$ were found to be:

$$(7) \quad \frac{\partial g}{\partial \alpha} = \int_{t=-\infty}^{\infty} (e^t - X_i - \alpha) A_i(t) dt / \tau^2$$

$$\frac{\partial g}{\partial \xi} = \int_{t=-\infty}^{\infty} (t - \xi) A_i(t) dt / \sigma^2$$

$$\frac{\partial g}{\partial \sigma^2} = \int_{t=-\infty}^{\infty} \{(t - \xi)^2 - \sigma^2\} A_i(t) dt / 2\sigma^4$$

$$\frac{\partial g}{\partial \tau^2} = \int_{t=-\infty}^{\infty} \{(e^t - X_i - \alpha)^2 - \tau^2\} A_i(t) dt / 2\tau^4$$

$$\frac{\partial^2 g}{\partial \alpha^2} = \int_{t=-\infty}^{\infty} \{(e^t - X_i - \alpha)^2 - \tau^2\} A_i(t) dt / \tau^4$$

$$\frac{\partial^2 g}{\partial \alpha \partial \xi} = \int_{t=-\infty}^{\infty} (t - \xi)(e^t - X_i - \alpha) A_i(t) dt / \sigma^2 \tau^2$$

$$\frac{\partial^2 g}{\partial \xi^2} = \int_{t=-\infty}^{\infty} \{(t - \xi)^2 - \sigma^2\} A_i(t) dt / \sigma^4$$

$$\frac{\partial^2 g}{\partial \alpha \partial \sigma^2} = \int_{t=-\infty}^{\infty} (e^t - X_i - \alpha) \{(t - \xi)^2 - \sigma^2\} A_i(t) dt / 2\sigma^4 \tau^2$$

$$\frac{\partial^2 g}{\partial \xi \partial \sigma^2} = \int_{t=-\infty}^{\infty} (t - \xi) \{(t - \xi)^2 - 3\sigma^2\} A_i(t) dt / 2\sigma^6$$

$$\frac{\partial^2 g}{\partial(\sigma^2)^2} = \int_{t=-\infty}^{\infty} \{(t - \xi)^4 - 6\sigma^2(t - \xi)^2 + 3\sigma^4\} A_1(t) dt / 4\sigma^8$$

$$\frac{\partial^2 g}{\partial\alpha\partial\tau^2} = \int_{t=-\infty}^{\infty} (e^t - X_1 - \alpha) \{(e^t - X_1 - \alpha)^2 - 3\tau^2\} A_1(t) dt / 2\tau^6$$

$$\frac{\partial^2 g}{\partial\xi\partial\tau^2} = \int_{t=-\infty}^{\infty} (t - \xi) \{(e^t - X_1 - \alpha)^2 - \tau^2\} A_1(t) dt / 2\tau^4\sigma^2$$

$$\frac{\partial^2 g}{\partial\sigma^2\partial\tau^2} = \int_{t=-\infty}^{\infty} \{(t - \xi) - \sigma^2\} \{(e^t - X_1 - \alpha)^2 - \tau^2\} A_1(t) dt / 4\tau^4\sigma^4$$

$$\frac{\partial^2 g}{\partial\tau^2} = \int_{t=-\infty}^{\infty} \{(e^t - X_1 - \alpha)^4 - 6\tau^2(e^t - X_1 - \alpha)^2 + 3\tau^4\} A_1(t) dt / 4\tau^8$$

Although in our application we will assume τ known, for the sake of completeness the list above includes the derivatives of g with respect to τ^2 as well as the other three parameters.

From these expressions, the derivatives of the log likelihood function follow at once, and these in turn may be used to compute maximum likelihood estimators (MLE's) of α , ξ and σ iteratively. If a second-order method is used to maximize the log likelihood, then the observed Fisher information matrix is obtained as a by-product, and can be used to obtain approximate (asymptotic) standard errors for the parameter estimates.

There is an important difference between the LNN and the 3PLND — unlike the 3PLND, the likelihood of the LNN has no singularities and is infinitely differentiable with respect to all of its parameters. In this connection, it is of interest to note that one of the methods used to avoid the problem of the singularity of the 3PLND (see for example Griffiths 1980) consists of a recognition that the observed values always involve some error,

even if it be only recording to a finite number of decimals, and recognition of this fact leads to a modified likelihood equation for the 3PLND in which the singularity is not present.

The LNN shares with the 3PLND another form of degeneracy however – that in which the data appear to conform better to the normal than the LNN distribution. When data of this type are subjected to maximum likelihood estimation, the successive estimates diverge, with $\alpha \rightarrow \infty$, $\sigma \rightarrow 0$, and ξ ‘taking up the slack’, implicitly fitting the normal distribution as the degenerate asymptotic LNN (see equation 6). Klimko et al. (1975) describe this problem as tending to arise in 3PLND data when the sample has a negative skewness, but a more precise analysis would be along the following lines:

Take a 3PLND sample – for example the underlying Z_i of this paper.

Assume α given and write

$$S_k = \sum_{i=1}^n [\ln(Z_i + \alpha)]^k / n.$$

Then the 3PLND likelihood maximized over ξ and σ may be written

$$(8) \quad 2 \ln L = -n\{1 + \ln(2\pi)\} - 2S_1 - \ln(S_2 - S_1^2).$$

A necessary condition for the MLE to be degenerate is that this quantity is increasing as $\alpha \rightarrow \infty$. Differentiating with respect to α , we find that this is true if

$$(9) \quad \sum_{i=1}^n \frac{1}{Z_i + \alpha} [\ln(Z_i + \alpha) - S_1 + S_2 - S_1^2] < 0 \text{ for large } \alpha.$$

A diagnostic for degeneracy of the 3PLND is therefore to compute (9) for some very large value of α . If its value is negative, it shows that there is at least a local degenerate solution; if it is positive, it proves that the solution is not degenerate.

Klimko et al. use the sign of the sample third moment as an indicator of likely degeneracy, but show that this is not always a reliable indicator. Evaluating third moment and (9) in a variety of test cases has shown that while in the majority of cases the two quantities have the same sign, on occasion they do not. These cases are samples in which Klimko's test would wrongly lead one to expect a degenerate solution while the diagnostic (9) would give the correct conclusion. We recommend the use of the more reliable criterion (9) to indicate 3PLND degeneracy instead of the skewness.

While this degeneracy criterion applies strictly only to the 3PLND, we have found it to be a reliable indicator of degeneracy in the LNN as well. That is, we redefine

$$S_k = \frac{1}{n} \sum_{i=1}^n [\ln(X_i + \alpha)^k] / n, \text{ and compute}$$

$$(10) \quad \frac{1}{n} \sum_{i=1}^n \frac{1}{X_i + \alpha} [\ln(X_i + \alpha) - S_1 + S_2 - S_1^2]$$

for a suitably large value of α , predicting degeneracy if its value is negative, and regularity otherwise. In practice, we have applied this test by using a trial value of α two or three orders of magnitude bigger than the largest order statistic of the data, taking its sign to be indicative of convergence or degeneracy. While the validity of this test is based only on the heuristic argument of analogy with the underlying 3PLND variable, to date we have found no data set in which it was wrong about convergence of maximum likelihood – either in predicting divergence where the estimation procedure converged, or in predicting convergence where the estimation procedure diverged.

It may be noted that the circumstances in which the LNN degenerates to the normal correspond to those of Cheng and Iles' (1990) 'embedded model' situation. An alternative possibility to our heuristic diagnostic is the development of a test for the embedded model based on an asymptotic expansion of the likelihood function in terms of α .

Methods of moments

While the method of maximum likelihood provides the theoretically ideal solution to the problem of parameter estimation, it has the practical drawback of involving very heavy computations because of the difficult quadratures required to evaluate all the quantities needed. This drawback creates interest in other possible estimation procedures that do not involve substantial calculations, but which may in at least some useful circumstances lead to acceptably accurate estimates of the parameters.

Analogously with the 3PLND, we start with method of moments estimators. The key to the moment estimators lies in the fact that the first and third central moments of X are identical to those of Z , while the variance of X exceeds that of Z by τ^2 . Thus, defining

$$\bar{X} = \sum_{i=1}^n X_i / n$$

$$s^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / n - \tau^2$$

$$m_3 = \sum_{i=1}^n (X_i - \bar{X})^3 / n$$

$$b_1 = m_3^2 / s^6$$

we obtain estimates of the first three moments of the underlying 3PLND variable Z , and are able to apply the moment equations of the 3PLND, whose solution (Johnson and Kotz 1970, p. 124) is:

$$(11) \quad \omega = \sqrt[3]{1 + \frac{1}{2}b_1 + D} + \sqrt[3]{1 + \frac{1}{2}b_1 - D} - 1$$

$$\hat{\sigma}^2 = \ln \omega$$

$$\hat{\xi} = \frac{1}{2} \ln[s^2 / \{\omega(\omega - 1)\}]$$

$$\hat{\alpha} = \exp(\hat{\xi} + \frac{1}{2}\hat{\sigma}^2) - \bar{X}$$

$$\text{where } D = \{(1 + \frac{1}{2}b_1)^2 - 1\}^{\frac{1}{2}}.$$

While the moment estimators for the LNN are very similar to those of the 3PLND, there is one noteworthy difference: s^2 is not the variance of the data X , but the variance less the counting variance τ^2 . It is thus possible (unlike the 3PLND case) for s^2 to be negative and consequently for the procedure to fail. When this occurs, it is an indication that ξ and/or σ is small; Z effectively constant; and that the LNN is primarily just measurement error.

As in the 3PLND case, the third moment of the LNN is necessarily strictly positive. Thus the method of moments estimator should be regarded as having failed if a data set has a negative third moment. When this is the case, it should be considered an indication that the data set conforms better to the normal distribution than the LNN; the normal is itself a degenerate case of the LNN.

The attraction of these moment estimators of the parameters is that they are in a computationally simple closed form, are statistically consistent, and are very easily computed. Their major disadvantage is that since the LNN distribution is very heavy tailed, they can be expected to be statistically quite inefficient. Nevertheless, particularly where large samples are available, the lack of efficiency may be acceptable while the computational load of the method of maximum likelihood (which increases with sample size) may make the method of moments estimators relatively attractive.

An unattractive feature of the method of moments estimator is its use of the third sample moment, a statistic which is likely to be poorly estimated in samples of moderate size because of the very heavy tail of the LNN distribution. To ameliorate the corresponding problem in the 3PLND distribution Cohen and Whitten (1980) proposed modified method of moments estimators, in which the equation based on skewness is replaced by some other equation. Particularly simple and effective is the system obtained by matching a sample quantile instead of the sample skewness, which suggests the adaptation of the same idea to the LNN distribution.

We have investigated this modified method of moments estimator. In view of the superiority of the modification in the 3PLND case, it might be expected that the modified method would be similarly advantageous for the LNN, but this turns out not to be the case. We have found it to be substantially less effective than the conventional three-moment estimator.

Comparison of the estimation procedures

A simulation experiment on the canonical LNN $\alpha = 0$, $\tau = 1$ was run to compare the estimation procedures. Six (ξ, σ) pairs were selected and 50 random samples of size 100 generated for each. The summary statistics of the resulting method of moments (MM) and maximum likelihood (ML) estimates are given in Table 1.

Several features of the table are noteworthy. First we see that both methods fail on occasion – the moment estimator by producing imaginary estimates, and maximum likelihood by producing the degenerate solution with $\sigma^2 = 0$ (implying that the normal distribution is fitted). Failures of ML are most common when ξ and/or σ are small.

The most striking feature of this table is the substantial bias of the estimates given by the method of moments. Clearly (at least for samples of size 100) this method is not attractive. Using it to provide starting values for maximum likelihood is very useful though; convergence to the MLEs starting from the MM estimates is usually quick, typically occurring in 10 or

fewer iterations.

Maximum likelihood shows little evidence of substantial bias, even though the sample size of 100 used is not of a size where one would normally invoke asymptotic consistency very confidently. It is thus to be recommended except perhaps where very large samples make the computational burden of ML substantial.

The method of moments seems unusable in its own right for small samples, but perhaps in large samples its consistency would remove enough of its bias to make it usable. We checked the moment estimators (but not, because of the computational load this would create, maximum likelihood) for random samples of size 5000, with the results listed in table 2. This table shows that the method of moments estimator was still somewhat biased though much less so than with small samples. It thus appears that if large samples are available, the method of moments may be used.

As regards the sample size simulated, in many contexts it might be objected that samples of size 5000 are unheard of, but samples of this size and even much larger are frequently found in gold mines.

5. INFERENCE ON Z FOR GIVEN X

The work on the LNN distribution was motivated by the framework in which there is interest in an underlying 3PLND variate which is observed with appreciable random error. A major concern in this situation is to estimate the underlying Z from the reading X . For this we require the conditional distribution of Z given X , as well as the moments of this distribution. At this point we return to the full four-parameter form of the distribution, but will work for notational convenience with $Y = Z + \alpha$ rather than Z , which can be recovered from Y very easily by subtracting α .

The conditional distribution of $Y|X = x$ is, by definition, $f(x, y)/g(x|\alpha, \tau, \xi, \sigma)$

$$= \frac{1}{2\pi\sigma\tau y} \exp - \frac{1}{2} \left[\frac{(\ln y - \xi)^2}{\sigma^2} + \frac{(x - y + \alpha)^2}{\tau^2} \right] / g(x|\alpha, \tau, \xi, \sigma)$$

While this distribution is not a standard form, its moments can be found quite easily, the k -th moment being given by:

$$E[Y^k | X = x] = \int_{y=0}^{\infty} y^k f(x, y) dy / g(x | \alpha, \tau, \xi, \sigma).$$

Writing

$$\begin{aligned} \int_{y=0}^{\infty} y^k f(x, y) dy &= \int_{y=0}^{\infty} \frac{y^k}{2\pi\sigma\tau} \exp - \frac{1}{2} \left[\left(\frac{\ln y - \xi}{\sigma} \right)^2 + \left(\frac{x + \alpha - y}{\tau} \right)^2 \right] dy \\ &= \int_{t=-\infty}^{\infty} \frac{e^{kt}}{2\pi\sigma\tau} \exp - \frac{1}{2} \left[\frac{(t - \xi)^2}{\sigma^2} + \frac{(x - e^t + \alpha)^2}{\tau^2} \right] dt \end{aligned}$$

and incorporating the $\exp(kt)$ term into the first term in square brackets and simplifying gives for this integral

$$e^{k\xi + \frac{1}{2}k^2\sigma^2} \int_{t=-\infty}^{\infty} \frac{1}{2\pi\sigma\tau} \exp - \frac{1}{2} \left[\frac{(t - \xi - k\sigma^2)^2}{\sigma^2} + \frac{(x - e^t + \alpha)^2}{\tau^2} \right] dt.$$

The integral in this expression can be recognized from the defining equations of the LNN distribution, and so we may write

$$(12) \quad E[Y^k | X = x] = e^{k\xi + \frac{1}{2}k^2\sigma^2} \frac{g(x | \alpha, \tau, \xi + k\sigma^2, \sigma)}{g(x | \alpha, \tau, \xi, \sigma)}.$$

The $\exp(k\xi + \frac{1}{2}k^2\sigma^2)$ term is the marginal k -th moment of the 2PLND variate Y , so that the k -th conditional moment may be thought of as the marginal k -th moment of Y multiplied by a conditioning correction factor – the ratio of the two g -terms.

The first two moments are sufficient to solve the calibration problem. The best estimate of Y given X is $E(Y|X)$, and the variance of prediction is $E(Y^2|X) - [E(Y|X)]^2$. As $Z = Y - \alpha$, this provides the answer to

the problem of point estimation of Z , with standard error, from its LNN reading.

Approximating the conditional distribution of Y given X

While the mean and standard error suffice for many purposes, it is also helpful, particularly if we seek interval estimates for Z , to have the conditional distribution of $Y|X$. For this, consider the formal expansion

$$\ln g(x|\alpha, \tau, \xi + \delta, \sigma) = \ln g(x|\alpha, \tau, \xi, \sigma) + a\delta + \frac{1}{2}b\delta^2 + \dots$$

Then

$$\begin{aligned} \ln E\{Y^k|X=x\} &= k\xi + \frac{1}{2}k^2\sigma^2 + ak\sigma^2 + \frac{1}{2}bk^2\sigma^4 + \dots \\ &= k(\xi + a\sigma^2) + \frac{1}{2}k^2(\sigma^2 + b\sigma^4) + \dots \end{aligned}$$

Now the k -th moment of 2PLND with logarithmic mean and variance ξ and σ^2 is $\exp(k\xi + \frac{1}{2}k^2\sigma^2)$, so this form of the natural log of $E\{Y^k|X=x\}$ suggests that to the extent that $\ln g$ can be approximated by a quadratic in ξ , the conditional distribution of Y given X may be approximated by a 2PLND and therefore that of Z by a 3PLND. As we shall show in section 7 by a numerical example, a lognormal approximation does indeed appear to fit well.

6. COMPUTATIONAL DETAILS

Returning to the convenient canonical form $\alpha = 0$, $\tau = 1$ in terms of which both g and its derivatives may be expressed, we find that g and its derivatives involve the terms

$$A(t) = \frac{1}{2\pi\sigma} \exp - \frac{1}{2} \left[\frac{t - \xi}{\sigma} \right]^2 \times \exp - \frac{1}{2} (e^t - x)^2$$

while G involves the term

$$\exp - \frac{1}{2} \left[\frac{t - \xi}{\sigma} \right]^2 \Phi(x - e^t).$$

The first term of each of these pairs tends to zero as $t \rightarrow \pm\infty$. The second term tends to zero very rapidly as $t \rightarrow \infty$, but tends slowly to a nonzero limit as $t \rightarrow -\infty$. Attempts were made to obtain better conditioned integrands using repeated integration by parts based on the iterated integrals of the error function (Abramowitz and Stegun 1965); but these attempts actually worsened the situation, and finally a direct evaluation using Romberg quadrature (Henrici 1964) was used. As the high-order derivatives of the integrand can vary enormously over the range of integration, we found it helpful to perform separate integrations in three subranges – a narrow central part where the high-order derivatives were large, and separate left and right tails. This provided a satisfactory algorithm with an internal estimate of accuracy and acceptable, though long, execution times.

Maximum likelihood estimation is the most time-consuming routine operation with LNN data since it involves many evaluations of g and its derivatives for each value in the sample. The initial estimates for maximum likelihood estimation are obtained from the method of moments. Where the data show little skewness (indicating that the solution, though not degenerate, may be quite close to degenerate), the initial α may be large and the initial σ near zero. Not only can this produce slow convergence, but the small σ can lead to numeric overflow problems. It is sometimes helpful to edit these starting values to produce starting values of σ sufficiently far above zero to avoid these numeric problems and speed up the convergence to the MLEs.

We have had good results applying maximum likelihood using a hybrid algorithm which moves smoothly between steepest-descent and Newton steps, behaving like steepest descent in the initial steps and like Newton's method as it nears the optimum.

7. EXAMPLE

As an illustration of the fitting and use of the distribution, we consider the following set of data obtained by counting a rich gold reef using the XRF portable gold analyzer (Lloyd 1980). Class midpoints (measured in centimetre-gram per tonne, or cm g/t) and frequencies were as follows:

X_i	f_i	X_i	f_i	X_i	f_i
-750	22	3250	167	10000	65
-250	188	3750	141	12000	40
250	639	4250	117	14000	16
750	649	4750	97	15000	10
1250	474	5500	152	16400	10
1750	322	6500	91	18500	10
2250	267	8000	95	28700	10
2750	226				

In an unpublished report by Professor H.S. Sichel, theoretical calculations using Poisson counting statistics of the instrument showed that there is some heteroscedasticity in the errors, but that this becomes apparent only at very large values of Z . For the remainder of the range, the constant value $\tau = 290$ fits well. The data show clearly the phenomenon of negative readings for gold, the lowest being approximately -1000 cm g/t. These negative values for a necessarily positive quantity result from the zero-mean normal measurement error, and are statistically to be expected though they cause much consternation in mining personnel not familiar with the characteristics of the instrument.

Figure 1 shows the result of plotting the cumulative distribution on a lognormal probability scale. For this plot, the non-positive values are included in the observation count, but are not plotted (their abscissae being imaginary). The plot is quite straight for normal scores above 1, and fitting a line by eye to this right portion of the plot gives the estimates 7.5 and 0.9 for ξ and σ respectively.

The method of moments gives the estimates:

$$\alpha = 1128, \xi = 7,90, \sigma^2 = 0,56.$$

Starting from the method of moments estimates, the maximum likelihood estimates, obtained in seven iterations, were:

$$\hat{\alpha} = 345, \hat{\xi} = 7,48, \hat{\sigma}^2 = 0,93.$$

The observed Fisher information gives the following approximate (asymptotic) standard errors for the parameters and correlation matrix between estimates:

	s.e.	Correlation		
α	24,1	1,00		
ξ	0,022	0,85	1,00	
σ^2	0,032	-0,83	-0,75	1,00

There is a tradeoff when using the XRF gold analyzer between cost and precision. Instead of measuring the gold face with the analyzer, one could use the conventional method of chipping samples from the face and performing a chemical assay on them. Doing so measures Z effectively exactly, but costs more and takes longer. A question of some interest then is how severely the measurement error in X impacts the quality of the parameter estimates and how much better would the estimates of the parameters be if the underlying Z had been observed without error. This question can be addressed by computing the expected Fisher information from a lognormal sample of the same size and with the same parameter values – in effect mimicking what one would obtain if one were to measure the underlying true lognormal variables Z directly. This was done using as the parameter values the ML estimates from the LNN fit, and gave the following standard errors and correlation matrix:

	s.e.	Correlation		
α	12,30	1,00		
ξ	0,019	0,58	1,00	
σ^2	0,029	-0,69	-0,40	1,00

A comparison of these two sets of figures shows that the measurement error has a large impact on the quality of the estimate of α , where the standard error is double what would be obtained if the underlying lognormal Z were measured directly. The effect on the other parameters is not as severe – the standard errors for the parameters ξ and σ^2 increase by 15% and 10% respectively.

The correlations between the parameter estimates are high, indicating that there is a range within which one can simultaneously adjust α , ξ and σ^2 giving a series of values fitting the data about equally well. This phenomenon is well known in the underlying 3PLND distribution of Z (Johnson and Kotz 1970, p. 122), and the presence of the measurement error exacerbates it.

In the mining application, the next objective would be the calibration curve $E(Z|X)$. This curve would be used to turn future instrument readings into conditionally unbiased estimates of the true grade at that point on the face. We get this calibration curve easily from the material of section 5, which provides expressions for $E(Y|X)$ and $\text{var}(Y|X)$, and from which we can obtain both a calibrated reading and its standard error.

While the standard error provides a measure of the uncertainty in Y (and hence Z) given X , the whole conditional distribution of Y given X is also of interest, as is the question of how well the conditional distribution of Y given X could be approximated for each X by a 2PLND. A partial answer to this is obtained by using the conditional mean and variance given in equation (12) to define a (notional) conditional logarithmic mean ξ_x and variance σ_x^2 implicitly from the equations:

$$E(Y|X = x) = \exp(\xi_x + \frac{1}{2}\sigma_x^2); \quad E(Y^2|X = x) = \exp[2(\xi_x + \sigma_x^2)].$$

It was suggested that the 2PLND be used to approximate the conditional distribution of Y given X . We may measure the quality of this approximation by comparing the exact third and fourth moments of the

conditional distribution of Y given X with the values $-\exp(3\xi_x + 4\frac{1}{2}\sigma_x^2)$ and $\exp(4\xi_x + 8\sigma_x^2)$ respectively – implied by the 2PLND approximation.

Table 3 shows the result of these various computations. It gives for each class midpoint X used in the data (i) the calibration curve predicting Z from X and the standard error of the predicted Z ; (ii) the computed conditional logarithmic mean and variance of Y given X ; and (iii) the natural log of the ratio of the exact third and fourth conditional moments to their 2PLND approximations.

Several features of this calibration deserve comment.

- (i) The calibration curve is nonlinear, though as $x \rightarrow \infty$, $E(Z|X=x) \rightarrow x$. By contrast, the regression of X on Z is a 45 degree line through the origin.
- (ii) The conditional logarithmic variance decreases as x increases, showing that in percentage terms the estimation of Z is much more precise at higher readings than at lower. The standard error on the linear scale is low for small X but increases toward and approximately equals the measurement standard deviation τ as X increases. The reason for this is that where X is near or below zero, considerable use is made of the 'prior' information about the distribution of Z , but where X is large, the measurement information predominates.
- (iii) Judged by the correspondence between the exact and approximate third and fourth moments, the 2PLND approximation for Y is excellent for moderate to large x and quite good for small x . This fact suggests that good confidence intervals for Z may be based on the approximation that for given x

$$\ln(Z + \alpha) \sim N(\xi_x, \sigma_x^2).$$

- (iv) A feature of the XRF analyzer that caused resistance by many potential users was its negative readings of grade. The table entry for $X = -750$ shows that even when the X -values are calibrated, negative estimates of grade remain possible. In fact as $x \rightarrow -\infty$, $E(Z|X = x) \rightarrow -\alpha$, which for this data set is estimated as -345 . In reality this is a reflection, not of the quality of the LNN distribution, but on the fact that the 3PLND model used for Z is itself erroneous in permitting (albeit with low probability) negative grades.

8. CONCLUSION

One of the author's favourite quotations was heard from B H P Rivett. It is to the effect that Babylonian astronomy, by using a vast accumulation of individually imprecise measurements, was able to make predictions whose quality was not matched by post-Galilean telescropy until about a century ago. Similar considerations apply to the trade-off between small numbers of effectively error-free measurements of underlying three-parameter lognormal variables and larger numbers of indirect measurements with substantial random measurement errors. Clearly enough of the latter will do a better job than the former. This situation may be arising in precious metal mining, and in environmental monitoring. For the low concentration analytes seen in both these applications, the 3PLND is a natural modeling distribution, and under some circumstances, homoscedastic normal measurement error may also be adequate.

This paper addresses some of the technical statistical issues that arise with this convolution of lognormal and normal distributions. An application to gold mining appeared in Magri and Hawkins (1987). It is an interesting question for the future whether there will be growth in measurements following this distribution.

Acknowledgements

The initial theoretical developments in this paper were made under the aegis of the National Research Institute for Mathematical Sciences, Council for Scientific and Industrial Research, Pretoria. The computations were carried out under Research Support Program RSP 1027 of the International Business Machines Corporation. The author gratefully acknowledges IBM's computing support; the permission of the Chamber of Mines Research Organization to make use of data and confidential reports by himself and Professor H.S. Sichel; and helpful discussions on numerical analysis with Dr D.P. Laurie; and the suggestions and careful checking of the referees.

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Table 1. Performance of the estimators

$\xi \sigma^2$	α		ξ		σ^2		Number
	Mean	sd	Mean	sd	Mean	sd	successful
<u>n = 100</u>							
-1 $\frac{1}{2}$ MM	0,203	0,589	-1,016	1,240	0,743	0,616	24
ML	0,230	0,315	-0,584	0,500	0,500	0,303	24
-1 1 MM	0,631	0,841	-0,323	1,191	0,580	0,611	46
ML	0,329	0,572	-0,663	1,023	0,989	1,059	38
-1 2 MM	0,859	0,499	0,197	0,454	0,712	0,198	50
ML	0,090	0,332	-0,884	0,764	1,987	1,440	50
0 $\frac{1}{2}$ MM	3,277	15,704	0,498	0,927	0,376	0,246	48
ML	0,097	0,669	-0,040	0,622	0,658	0,482	40
0 1 MM	0,955	0,614	0,691	0,281	0,536	0,264	50
ML	0,177	0,675	0,069	0,528	1,076	0,590	50
0 2 MM	2,446	0,952	1,263	0,243	0,689	0,225	50
ML	0,022	0,299	0,004	0,379	2,081	0,811	50
5 $\frac{1}{2}$ MM	67,866	43,118	5,398	0,207	0,303	0,132	50
ML	-6,444	13,177	4,948	0,133	0,563	0,148	50
5 1 MM	128,182	50,005	5,672	0,194	0,484	0,172	50
MM	-5,644	7,642	4,953	0,118	1,112	0,219	50
5 2 MM	341,174	164,314	6,154	0,307	0,774	0,254	50
ML	-1,923	3,956	4,935	0,156	2,097	0,388	50

Table 2: Large-sample performance of the MM estimator

$\xi \sigma^2$	Mean	α sd	Mean	ξ sd	Mean	σ^2 sd	Number successful
<u>n = 5000</u>							
0 $\frac{1}{2}$	0,022	0,138	0,017	0,147	0,495	0,094	50
0 1	0,169	0,211	0,147	0,225	0,892	0,210	50
0 2	0,995	0,332	0,575	0,176	1,486	0,292	50
-1 $\frac{1}{2}$	0,050	0,151	-0,936	0,377	0,492	0,178	50
-1 1	0,055	0,088	-0,875	0,229	0,907	0,203	50
-1 2	0,382	0,122	-0,383	0,218	1,407	0,279	50
5 $\frac{1}{2}$	5,087	14,950	5,033	0,115	0,488	0,076	50
5 1	24,486	25,443	5,147	0,175	0,897	0,163	50
5 2	134,154	44,726	5,540	0,223	1,481	0,304	50

Table 3. Approximation of posterior distribution for Z

Calibration curve				2PLND approximation to distribution			
x Predicted Z se				ξ_x	σ_x^2	Error in	
						μ_3	μ_4
1	-750	-86,09	125	5,4450	0,4598	-0,0422	-0,1568
2	-250	47,41	177	5,8753	0,4317	-0,0422	-0,1546
3	250	310,45	241	6,4192	0,3572	-0,0287	-0,1043
4	750	723,71	279	6,9398	0,2570	-0,0106	-0,0392
5	1250	1205,54	288	7,3284	0,1844	-0,0032	-0,0122
6	1750	1703,10	290	7,6140	0,1410	-0,0011	-0,0044
7	2250	2204,28	290	7,8365	0,1137	-0,0005	-0,0019
8	2750	2706,38	290	8,0183	0,0950	-0,0002	-0,0009
9	3250	3208,62	290	8,1720	0,0817	-0,0001	-0,0005
10	3750	3710,77	290	8,3050	0,0716	-0,0001	-0,0003
11	4250	4212,77	290	8,4222	0,0637	0,0000	-0,0002
12	4750	4714,59	290	8,5271	0,0574	-0,0000	-0,0001
13	5500	5467,01	290	8,6662	0,0499	-0,0000	-0,0001
14	6500	6469,77	290	8,8257	0,0426	-0,0000	-0,0000
15	8000	7973,04	290	9,0254	0,0349	-0,0000	-0,0000
16	10000	9976,38	290	9,2414	0,0281	-0,0000	-0,0000
17	12000	11978,92	290	9,4189	0,0235	-0,0000	-0,0000
18	14000	13980,91	290	9,5695	0,0203	-0,0000	-0,0000
19	15000	14981,76	290	9,6371	0,0189	0,0000	0,0000
20	16400	16382,81	290	9,7246	0,0173	-0,0000	-0,0000
21	18500	18484,16	290	9,8430	0,0154	-0,0000	-0,0000
22	28700	28688,38	290	10,2761	0,0100	-0,0000	-0,0000

Fig 1. Log probability plot of sample data

